LAB ACTUARIAL
MATHEMATICS

CLAIMS RESERVING

CHAIN LADDER VS. BF ON PRIVATE AUTO DATA

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PROJECT OVERVIEW

To estimate the outstanding claims reserve for a private passenger auto insurance portfolio using two actuarial methods, Chain Ladder and Bornhuetter-Ferguson (BF), and to compare their forecasts with actual outcomes.

Data

- Source: CAS Loss Reserving Database (NAIC Schedule P)
- Insurer: State Farm Mut Grp (GRCODE 1767)
- Line of business: Private Passenger Auto
- Triangle: cumulative paid claims, accident years 1988–1991, 4 development years
- Amounts in thousands of USD
- Inflation: CPI-U annual averages (FRED, series CPIAUCNS)

Methodology

1. Build the cumulative paid loss triangle
2. Estimate development factors and project the Chain Ladder reserve
3. Adjust payments to 1991 prices using CPI-U and recompute the reserve
4. Compute the BF reserve using an expected loss ratio
5. Compare both methods against the actual outcomes (back-test)

RESULTS

Reserve Comparison (thousands of USD)

Accident Year	Chain Ladder	BF	Actual	CL Error	BF Error
1988	0	0	0	-	-
1989	606,294	597,811	601,780	+0.8%	-0.7%
1990	1,903,542	1,905,628	1,861,125	+2.3%	+2.4%
1991	4,796,479	5,081,266	4,505,840	+6.5%	+12.8%
Total	7,306,315	7,584,705	6,968,745	+4.8%	+8.8%

Totals may differ by 1 due to rounding.

Key Findings

- Both methods overestimated the total reserve. Chain Ladder was closer to the actual outcome (+4.8% vs. +8.8% for BF).
- Forecast error grows for more recent accident years, since they rely on more projected development.
- BF used an expected loss ratio of 0.799, taken from accident year 1988 alone. This likely explains its larger error in 1991.
- After adjusting payments to 1991 prices (CPI-U), the Chain Ladder reserve was 6,826,444. This is stated in 1991 prices, so it is not directly comparable with the nominal actual outcome.
- In 1991, paid claims after four years were only 72.7% of premium, well below the 79.9% assumed by BF.

CONCLUSION

Summary

- Chain Ladder estimated the total reserve at 7.31M, 4.8% above the actual outcome of 6.97M.
- Bornhuetter-Ferguson estimated 7.58M (+8.8%), driven mainly by accident year 1991 (+12.8%).
- In this dataset, Chain Ladder performed better than BF, mainly because BF depends on a single expected loss ratio taken from 1988.
- Restating payments in 1991 prices gave a Chain Ladder reserve of 6.83M (not directly comparable with the nominal figures above).

Limitations

- Only one insurer and one line of business were analysed.
- The triangle covers four years, while claims in the full data keep developing beyond year 4, so tail development is not captured.
- The loss ratio was based on one accident year.
- Annual average CPI-U was used as a simple approximation of claim inflation.

Possible Extensions

- Use the full 10×10 triangle to capture tail development.
- Test other insurers and lines of business.
- Estimate the expected loss ratio from several years.

APPENDIX

R CODE OUTPUTS, STEP BY STEP

- Step 1: Company selection. Top 10 insurers by 1988 net earned premium (thousands USD)

```
> big <- aggregate(EarnedPremNet_B ~ GRCODE + GRNAME,
+                  data = subset(ppauto,AccidentYear==1988 & DevelopmentLag==1),
+                  FUN=sum)#1988 اعلى 10 شركات من حيث اجمالي الاقساط في
> big <- big[order(-big$EarnedPremNet_B),]
> head(big,10)#ترتيب الشركات من الاكبر للاصغر
    GRCODE                              GRNAME EarnedPremNet_B
128   1767                  State Farm Mut Grp         7809394
133   2003 United Services Automobile Asn Grp         1106097
48    4839                  FL Farm Bureau Grp          258278
89    7080       New Jersey Manufacturers Grp           91247
43     388                  Federal Ins Co Grp           83473
86    3240             NC Farm Bureau Ins Grp           81870
68    1090   Kentucky Farm Bureau Mut Ins Grp           70105
130   6947                    Tenn Farmers Mut           63660
35     620     Employers Mut Co Of Des Moines           48731
114    965                       Secura Ins Co           33640
```

- Step 2: Cumulative paid loss triangle. Full 10×10 triangle, State Farm Mut Grp (GRCODE 1767)

```
> tri <- xtabs(CumPaidLoss_B ~ AccidentYear + DevelopmentLag,
+              data=co)
> tri
            DevelopmentLag
AccidentYear       1       2       3       4       5       6       7
        1988 2439272 4722902 5705646 6238289 6519491 6677426 6750431
        1989 2828267 5368026 6494597 7096377 7417869 7575814 7655217
        1990 3186948 5913490 7140613 7774615 8096374 8251086 8325184
        1991 3192619 5878000 7075254 7698459 7996404 8140065 8215810
        1992 3561950 6474291 7763969 8404713 8716926 8876813 8959761
        1993 3895076 7024867 8343417 9003120 9337099 9509203 9599735
        1994 4323103 7590944 8924062 9640098 10030681 10216964 10299231
        1995 4491070 7664190 9006113 9736570 10120120 10312088 10412421
        1996 4444088 7486113 8845221 9583622 9999739 10196271 10293776
        1997 4344144 7305064 8614474 9379418 9792901 9988209 10076219
            DevelopmentLag
AccidentYear       8       9      10
        1988 6787444 6808809 6815646
        1989 7693240 7712077 7721911
        1990 8364955 8382479 8394117
        1991 8256338 8279401 8288143
        1992 8998462 9016704 9026329
        1993 9642522 9660532 9673610
        1994 10340679 10361560 10375605
        1995 10461753 10497027 10512108
        1996 10347058 10373438 10387245
        1997 10123792 10148983 10165481
```

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- Step 3: Training triangle. 4×4 triangle with the lower part hidden as unknown (tri4_known)

```
            DevelopmentLag
AccidentYear       1       2       3       4
        1988 2439272 4722902 5705646 6238289
        1989 2828267 5368026 6494597 7096377
        1990 3186948 5913490 7140613 7774615
        1991 3192619 5878000 7075254 7698459
>
> tri4_known <- tri4
> tri4_known[row(tri4)+ col(tri4) >5] <-NA #مجهول السفلي الجزء نخلي NA
> tri4_known
            DevelopmentLag
AccidentYear       1       2       3       4
        1988 2439272 4722902 5705646 6238289
        1989 2828267 5368026 6494597      NA
        1990 3186948 5913490      NA      NA
        1991 3192619      NA      NA      NA
```

- Step 4: Development factors. Chain Ladder factors, years 1 to 4

```
> n <- 4
> f <- numeric(n-1)
> for(j in 1:(n-1)){
+   rows <- 1:(n-j)
+   f[j] <- sum(tri4_known[rows, j+1]) / sum(tri4_known[rows, j])
+ }
> round(f, 3)# العوامل
[1] 1.893 1.209 1.093
```

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R CODE OUTPUTS, STEP BY STEP

- Step 5: Chain Ladder projection and reserve. Completed triangle and reserve by accident year

```
> # a copy fill with expectation
> full <- tri4_known
>
> for(j in 2:n){
+   for(i in 1:n){
+     if(is.na(full[i,j])){
+       full[i,j]<- full[i,j-1] * f[j-1]
+     }
+   }
+ }#اذا فيه خانه غير معرفه نضرب الخانه الي قبلها في المعامل
> round(full)
            DevelopmentLag
AccidentYear       1       2       3       4
        1988 2439272 4722902 5705646 6238289
        1989 2828267 5368026 6494597 7100891
        1990 3186948 5913490 7149592 7817032
        1991 3192619 6043656 7306966 7989098
> latest <- tri4_known[cbind(1:n, n:1)]
> ultimate <- full[,n]
> reserve <- ultimate - latest
> round(reserve)#الاحتياطي لكل سنة حادث
   1988    1989    1990    1991
      0  606294 1903542 4796479
> sum(reserve)#الاحتياطي الكلي
[1] 7306316
```

- Step 6: Inflation adjustment. CPI-U rates and conversion factors to 1991 prices

```
> #from FRED CPIAUCNS we took inflation rates 1988-1991
> cpi <- c("1988"= 118.258, "1989"=123.967,
+          "1990"=130.658, "1991"=136.192)
> infl <- cpi[-1] / cpi[-4] - 1
> round(infl * 100 ,2)
1989 1990 1991
4.83 5.40 4.24
> idx <- cpi ["1991"] / cpi
> round(idx, 4)
  1988   1989   1990   1991
1.1517 1.0986 1.0424 1.0000
```

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R CODE OUTPUTS, STEP BY STEP

Step 7: Inflation-adjusted incremental and cumulative payments

```
> #1991 الدفعات بقيمة
> fac <- matrix(idx[as.character(cal)], n ,n)
> inc_adj <- inc * fac
> round(inc_adj)
            DevelopmentLag
AccidentYear       1       2       3      4
        1988 2809191 2508830 1024368 532643
        1989 3107176 2647330 1126571     NA
        1990 3321931 2726542      NA     NA
        1991 3192619      NA      NA     NA
>
> # نفس خطوات التوقع السابقه لكن باستخدام قيم ما بعد التضخم
> cum_adj <- t(apply(inc_adj, 1, cumsum))
> round(cum_adj)
            DevelopmentLag
AccidentYear       1       2       3       4
        1988 2809191 5318021 6342389 6875032
        1989 3107176 5754507 6881078      NA
        1990 3321931 6048473      NA      NA
        1991 3192619      NA      NA      NA
```

Step 8: Inflation-adjusted reserve (1991 prices)

```
> full_adj <- cum_adj
>
> for(j in 2:n){
+   for(i in 1:n){
+     if(is.na(full_adj[i,j])){
+       full_adj[i,j]<- full_adj[i,j-1] * f_adj[j-1]
+     }
+   }
+ }# تةوقع القيم باستخدام المعاملات والمبالغ بعد التضخم
> round(full_adj)
            DevelopmentLag
AccidentYear       1       2       3       4
        1988 2809191 5318021 6342389 6875032
        1989 3107176 5754507 6881078 7458960
        1990 3321931 6048473 7223443 7830079
        1991 3192619 5916764 7066149 7659575
>
> latest_adj <- cum_adj[cbind(1:n, n:1)]
> ultimate_adj <- full_adj[,n]
> reserve_adj <- ultimate_adj - latest_adj
> round(reserve_adj)
   1988    1989    1990    1991
      0  577883 1781606 4466956
> sum(reserve_adj)#1991 الاحتياطي الكلي بقيمة
[1] 6826444
```

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- Step 9: Bornhuetter-Ferguson. Premiums, expected loss ratio, and BF reserve

```
> prem <- sapply(1988:1991, function(y)
+   co$EarnedPremNet_B[co$AccidentYear == y & co$DevelopmentLag == 1])
> names(prem) <- 1988:1991
> prem #استخراج مبالغ الاقساط
    1988     1989     1990     1991
 7809394  8764863  9796463 10594952
>
> #ELR
> elr <- tri4[1, n] / prem["1988"]
> elr
     1988
0.7988186
>
> cdf <- sapply(1:n, function(k) if (k < n) prod(f[k:(n-1)])else 1)
> cdf_ay <- cdf[n:1]
> cdf_ay
[1] 1.000000 1.093354 1.321898 2.502365
>
> #BF
> bf <- prem * elr * (1 - 1 / cdf_ay)
> round(bf)
   1988    1989    1990    1991
      0  597811 1905628 5081266
> sum(bf)
[1] 7584705
```

- Step 10: Method comparison. Chain Ladder vs. BF vs. actual outcomes

```
> comp2 <- data.frame(
+   ChainLadder = round(reserve),
+   BF = round(bf),
+   Actual = actual_reserve
+   )
> comp2$CL_Err_Pct <- round(100 * (comp2$ChainLadder - comp2$Actual) / comp2$Actua
l, 1)
> comp2$BF_Err_Pct <- round(100 * (comp2$BF - comp2$Actual) / comp2$Actual, 1)
> comp2
     ChainLadder      BF  Actual CL_Err_Pct BF_Err_Pct
1988           0       0       0        NaN        NaN
1989      606294  597811  601780        0.8       -0.7
1990     1903542 1905628 1861125        2.3        2.4
1991     4796479 5081266 4505840        6.5       12.8
> colSums(comp2[,1:3])#المجاميع الكليه للاعمده الثلاث
ChainLadder          BF      Actual
    7306315     7584705     6968745
```